

Rudransh Goel

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Indian Institute of Technology Kanpur

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 RudranshGoel |  Rudransh Goel

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	9.30/10
2021	Class XII (CBSE)	Silver Line Prestige school	96.6%
2019	Class X (ICSE)	St. Paul's Academy	98.2%

Scholastic Achievements

- **Top 300**, among 2500+ participants, selected for Quant Trading Winter School by **Optiver & Eco. Club, IITD Jan '23**
- Received **Academic Excellence Award** by IIT Kanpur for exceptional academic performance *May '22 & December '22*
- **AIR 1091** in **Joint Entrance Examination (JEE) - Advanced**, among 150,000 candidates in India *October '21*
- **AIR 854** in **Joint Entrance Examination (JEE) - Mains**, among 1 million candidates in India *September '21*
- **AIR 889** in **Kishore Vaigyanik Protsahan Yojna (KVPY)**, SX stream, among 50,000 candidates *March '21*
- **Top 347** in **Indian Olympiad qualifier in Physics (IOQP)-I** among 20k+ candidates *February '21*
- **AIR 902** in **Kishore Vaigyanik Protsahan Yojna (KVPY)**, SA stream, among 50,000 candidates *December '19*

Research Experience

SURGE IITK | Mathematical Modelling, Parameter Estimation and Simulation of a Tracked Vehicle *May'23-July'23*

Faculty advisor: Prof. Ramprasad Potluri

- Extended **mathematical model** development for tracked vehicles, accounting for dynamically changing mass distribution
- Implemented the model in **MATLAB** for circular and straight-line movements and incorporated sloped terrain scenarios
- Performed simulations on **Simulink** to evaluate and refine the **DOB-based sensing** control of the tracked vehicle
- Used **Matlab's Sensitivity Analysis Toolbox** to evaluate the impact of parameter changes on vehicle's behavior
- Contributed to the **optimization** and **fine-tuning** of the tracked vehicle control system under various operating conditions

Key Projects

Mathematical Predictor: Machine Learning | Summer project mentor - Stamatics Society, IIT Kanpur | *May '23-July '23*

- Mentored a group of **50+** students in a learning project centered around gaining an understanding various ML concepts
- Developed and implemented a curriculum focusing on mathematical concepts applied in machine learning
- Designed **lecture material**, practical exercises and **assessments**, covering topics such as Linear Regression, Gradient Descent, Logistic Regression, Support Vector Machines, Ensemble Learning, Bayesian methods, and basic exposure to CNNs

Automated Chessboard | Summer project - Electronics Club, IIT Kanpur May '22-July '22

- Designed a **physical model** to automate chess game using stepper motors, reed switches, electromagnets, and **Arduino**
- Coded a **chess algorithm** involving the implementation of **Minimax algorithm** and **Alpha-Beta pruning**

Machine Learning- Behind the Curtain | Summer project- Stamatics Society, IIT Kanpur Apr'22-June'22

- Introduction to ML concepts such as linear and logistic regression, gradient descent, association rule learning, clustering
- Studied various techniques such as **hyperparameter tuning, regularization, and optimization techniques**

Positions of Responsibility

Senior Technical Team Member Team Vision, IIT Kanpur	May'23-Present
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- Guiding a team of **15+** students in training and participation for various competitions in autonomous ground navigation
- Aided in reintroduction of **hardware-based** competitions within the team, aligning with the team's pre-COVID era initiatives
- Revitalised the team by establishing **stronger roots** and set the momentum for future growth and exploration

Secretary | Electronics Club, IIT Kanpur Aug'22-Apr'23

- Conducted a 2-day workshop with **150+** participants and mentored them in projects such as Air Piano and Clap Switch
- Organized and managed the logistics of events, compilation of content, and gave lectures on introduction to C++

Assistant Editor | Vox Populi, IIT Kanpur Apr'22-May'23

- Led a team of **60+** CGMs in the publication of various articles for Vox Populi, the student journalism body of IIT Kanpur
- Reported water quality data from **50+** campus spots leading to discussion on water reforms by the Institute Administration
- Significantly expanded reader base to **17k** through the consistent publication of compelling and influential articles

Technical Skills

Programming Languages: C, C++, L^AT_EX, Python

Software and Libraries: Git, Numpy, Pandas, Matplotlib, Matlab

Relevant Courses

A*- *Exceptional Performance*

Probability and Statistics(A*)	Data Structures and Algorithms
Fundamentals of Computing	Linear Algebra and Ordinary Differential Equations
Real Analysis and Multivariable Calculus	Partial Differential Equations
Introduction to Electronics	Signals and Systems

Extra Curricular Activities

- **Secretary, Skating Club:** Taught skating to over **300** participants in multiple workshops, and promoted the sport
- Lead a team of **5** freshmen in **Interhall Technical Competition**, Takneek'22, working on a random number generators
- Attended the Vijyoshi Camp, annual national science camp, organised by **IISC, Bangalore** and **IISER, Kolkata**